

1. THEME & CHALLENGE AREAS

THE 2050 KERALA HOME

Imagine a home in Kerala that is affordable, low-carbon, climate-resilient and comfortable.

Participants are invited to envision what a typical Kerala home could look, function and perform like in 2050.

The proposed concept may address the building itself, its materials, systems, surroundings, construction process, operation, adaptability, or the way people interact with the home.

Solutions should demonstrate how a future Kerala home can balance:

Affordability + Sustainability + Resilience + Comfort + Practicality

Participants may explore one or more of the following challenge areas.

1.1 Affordable & Accessible Housing

How can sustainable homes become affordable and accessible to a wider section of society?

Possible areas include:

- Cost-effective housing systems
- Affordable sustainable materials
- Reduction of construction costs
- Locally available materials and resources
- Modular and prefabricated construction
- Efficient construction techniques
- Incremental housing
- Adaptable housing models
- Community-based housing approaches
- Reducing lifecycle and maintenance costs

1.2 Low-Carbon Materials & Construction

How can the carbon footprint of a Kerala home be reduced?

Possible areas include:

- Low-carbon construction materials
- Recycled and reused materials
- Bio-based materials
- Locally sourced materials
- Alternative building materials
- Reduction of embodied carbon
- Low-carbon structural systems
- Material-efficient construction
- Reuse of existing building components
- Innovative construction techniques
- Appropriate traditional and vernacular building practices

1.3 Climate-Responsive & Resilient Homes

How can homes respond to Kerala's changing climate?

Possible areas include:

- Flood-resilient housing
- Heat-resilient design
- Heavy-rainfall response
- Disaster-resilient construction
- Climate-responsive site planning
- Elevated or adaptable structures
- Water management
- Passive design strategies
- Natural ventilation
- Shading and solar control

- Thermal comfort
- Climate-adaptive local construction practices

1.4 Energy-Efficient & Comfortable Homes

How can the 2050 Kerala home remain comfortable while reducing energy consumption?

Possible areas include:

- Passive cooling
- Natural ventilation
- Daylighting
- Solar shading
- Thermal insulation
- Energy-efficient building envelopes
- Renewable energy integration
- Rooftop solar systems
- Energy-efficient appliances and systems
- Smart energy management
- Reduction of operational energy demand
- Thermal comfort without excessive dependence on mechanical cooling

1.5 Water-Positive & Resource-Efficient Homes

How can future homes use resources more responsibly?

Possible areas include:

- Rainwater harvesting
- Greywater reuse
- Water-efficient fixtures
- Household water management
- Decentralised wastewater treatment
- Resource-efficient construction

- Reduction of construction waste
- Material recovery and reuse
- Household-level waste segregation
- Circular resource systems
- Lifecycle-based resource management

1.6 Circular & Regenerative Homes

Can a home become part of a circular system rather than a source of continuous resource consumption?

Possible areas include:

- Reuse of construction materials
- Recycled materials
- Design for disassembly
- Construction and demolition waste reduction
- Waste-to-resource approaches
- Material recovery
- Circular construction systems
- Composting and organic waste management
- Productive landscapes
- Integration of natural systems into housing

1.7 Adaptable & Future-Ready Homes

What should a home designed for 2050 be capable of adapting to?

Participants may explore:

- Changing family structures
- Ageing populations
- Changing work patterns
- Work-from-home spaces

- Flexible room configurations
- Modular extensions
- Universal accessibility
- Multigenerational living
- Future technologies
- Changing climatic conditions
- Easy maintenance and modification
- Design for long-term adaptability

1.8 Smart, Digital & AI-Enabled Homes

How can technology make future homes more sustainable, efficient and responsive?

Possible areas include:

- AI-assisted home design
- Smart energy management
- Digital building performance monitoring
- Automated climate control
- Water monitoring
- Smart waste management
- Digital material selection
- Building performance simulation
- Predictive maintenance
- AI-assisted sustainability assessment
- Digital twins
- Technology-enabled resource optimisation

Technology should be used where it creates meaningful value and should remain appropriate to the economic and social context of Kerala.

1.9 Kerala's Vernacular Knowledge & Future Innovation

How can Kerala's traditional knowledge contribute to the home of 2050?

Participants may explore:

- Traditional Kerala architecture
- Climate-responsive vernacular practices
- Local materials
- Traditional passive cooling techniques
- Courtyards and shaded spaces
- Traditional water-management practices
- Indigenous construction knowledge
- Adaptation of traditional techniques using modern technology
- Combining vernacular wisdom with contemporary construction

The objective is not simply to reproduce traditional architecture, but to explore how relevant principles can inform future housing.

Participants may also propose solutions outside the above areas, provided the concept clearly contributes to the vision of The 2050 Kerala Home.

A concept may address a single challenge area or combine multiple aspects into an integrated housing solution.